

Turning “Unusable” Nail Images into a Condition Signal for Anemia Prediction

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ABSTRACT

This project began from a practical problem in nail-image anemia prediction: many images were removed because they contained nail polish, obstruction, pathology, or other unreliable visual conditions. Instead of treating these samples only as failed data, I tested whether one excluded condition could be learned as a warning signal for a future medical image model. After inspecting the excluded dataset, I found that most available excluded samples were polish-related, while obstruction and pathology had too few examples for reliable multi-class training. Therefore, I narrowed the final implementation to polish/no-polish classification. I separated usable and excluded images, generated nail-only crops using polygon annotations, applied group-based splitting, and fine-tuned a pretrained ResNet18 model. On the full-distribution test set of 595 nail crops, the classifier reached 98.7% accuracy, with no-polish recall of 1.00 and polish recall of 0.90. Because the test set was imbalanced, I interpreted the result using per-class recall and the confusion matrix instead of accuracy alone. This project does not complete anemia prediction, but it supports a first step toward a condition-aware anemia pipeline.

Keywords: anemia prediction; nail images; unusable data; polish detection; ResNet18

1. Introduction

This project started from a dataset problem in my nail-image anemia prediction work. Previous research has shown that fingernail images can support non-invasive anemia or hemoglobin-related screening because nail-bed color contains useful visual information (Mannino et al., 2018). More recent work also supports the use of skin and fingernail RGB images for hemoglobin assessment (Yakimov et al., 2024). However, real-world nail images are not always clean. Some contain nail polish, clear polish, obstruction, pathology, blur, or other visual problems. A normal pipeline removes these images before training, but this may hide an important issue: users will not always provide ideal nail photos.

From the actual data, I noticed that “unusable” did not always mean meaningless. Nail polish, for example, is a visible condition that changes the image but still has structure. This led to the main question of this project: can excluded nail images become a condition signal instead of only being discarded?

My original proposal was to build a multi-task anemia model with two outputs: one for anemia prediction and one for image-condition classification. After inspecting the dataset, I found that the excluded categories were not balanced enough for that full plan. Polish had enough samples, but obstruction and pathology had very few examples. Because of this, the final project focused on a smaller proof of concept: polish versus no-polish classification.

One inspiration for this topic was AlphaGo’s unexpected move against Lee Sedol, which first looked strange but later revealed a different strategic possibility through machine learning and search (Silver et al., 2016). In a smaller way, this project asks whether data that first looks unusable can still reveal a useful warning signal.

2. Methods

The project followed a bottom-to-top workflow: inspect the data, generate nail crops, construct the binary dataset, train the classifier, and interpret the result for a future anemia system.

First, I inspected the excluded image data. The excluded dataset contained 104 rows. Most were polish-related: 95 samples were labeled as polish. The remaining categories were much smaller, including obstruction, pathology, anomaly, and unknown cases. After organizing the files, the categorized folders contained 95 nail-polish images with corresponding JSON annotation files, only 3 obstructed images, and only 1 pathology image. This showed that a multi-class condition classifier would not be reliable for the final project.

Second, I separated raw PNG files into usable and excluded groups, producing 862 usable PNGs and 102 excluded PNGs. I then generated nail-only crops using polygon annotations from AnyLabeling-style JSON files. This step was necessary because a classifier trained on the whole image might learn background, skin, lighting, or camera clues instead of nail condition. The preprocessing notebooks exported 3508 usable nail-only crops and 402 polish nail-only crops.

Third, I built a polish/no-polish dataset using all 3910 available crops. Group-based splitting was used so crops from the same original image group would not appear in both training and testing. This reduced leakage and made evaluation more realistic. The final split contained 2722 training samples, 593 validation samples, and 595 test samples.

Lastly, I fine-tuned a pretrained ResNet18 classifier. ResNet introduced skip connections that help train deeper visual models (He et al., 2016). Since my dataset was limited, I used transfer learning instead of training a CNN from scratch. I froze most of the pretrained network and trained only the later convolutional block and final fully connected layer.

3. Results and Discussion

The final classifier was trained on 3910 nail-only crops: 3508 no-polish crops and 402 polish crops. The best validation accuracy was 0.9747. On the test set, the model reached 0.9866 accuracy.

Figure 1. Training and validation accuracy of the segmented nail polish classifier.

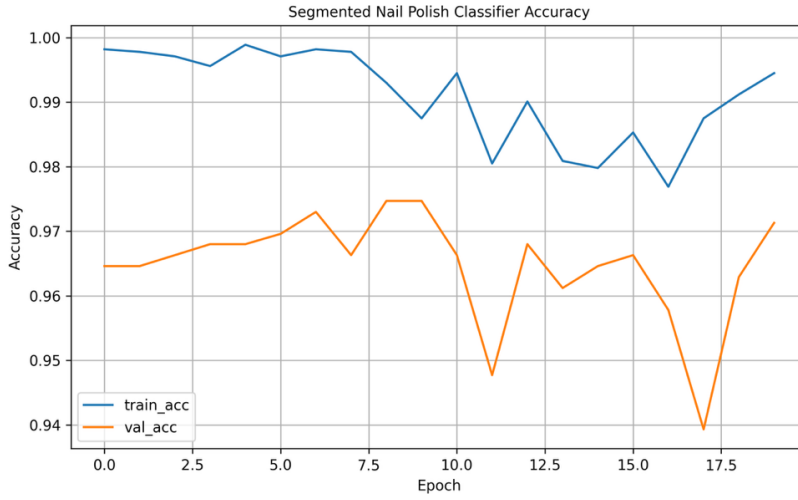


Table 1. Test-set result of the full-distribution model.

Class	Precision	Recall	F1-score	Support
No-polish	0.99	1.00	0.99	533
Polish	0.97	0.90	0.93	62
Macro average	0.98	0.95	0.96	595
Weighted average	0.99	0.99	0.99	595

The confusion matrix was $[[531, 2], [6, 56]]$. This means the model correctly classified 531 out of 533 no-polish crops and 56 out of 62 polish crops. Only 2 no-polish crops were falsely predicted as polish, while 6 polish crops were missed and predicted as no-polish.

This result supports the main idea that an excluded image condition can be learned from segmented nail crops. The model performed strongly on no-polish images, with recall of 1.00, and detected most polish crops, with recall of 0.90. However, the missed polish cases are important. In a future anemia prediction system, missed polish could cause the system to trust an image whose nail color has been visually altered.

The result should also be interpreted with the class distribution in mind. The test set contained many more no-polish crops than polish crops. Because of this imbalance, the high overall accuracy should not be the only evidence. Polish recall and macro-F1 are more useful for judging whether the minority condition is actually being learned.

The bottom-to-top process also changed the final project direction. At the proposal stage, I wanted a full multi-task anemia model. After checking the real dataset, I found that only polish had enough samples for a reliable condition classifier. This narrowing made the final result more honest: the model tested one excluded condition with a measurable result instead of pretending to solve all excluded categories poorly.

4. Conclusion

This project tested whether excluded nail images can become useful condition signals. I began with the actual dataset, found that polish was the only excluded category with enough samples, created nail-only crops from polygon annotations, and fine-tuned a ResNet18 classifier.

The final model reached 98.7% test accuracy, with polish recall of 0.90. This supports the idea that excluded nail conditions are not necessarily empty noise. They can contain learnable structure. At the same time, the model still missed 6 polish crops, so it is not yet reliable enough for medical deployment.

The practical implication is that a future anemia prediction system should not only predict a medical label. It should also understand whether the input image condition is trustworthy. A condition head could warn about polish, obstruction, pathology, or other visual interference before the anemia prediction is interpreted. In this project, the complete multi-task anemia model was not finished, but the first measurable step was completed: turning one type of unusable data into a learnable condition signal.

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